

AI Audit Report — HW03 Task 3: Cross-Browser / Cross-Platform

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0. This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

1. Student Information

Field	Value
Student name (printed):	DANG TRUONG NGUYEN
Student ID:	23127438
Class / Cohort:	23KTPM3
Assignment ID (e.g., HW#00, HW#02):	HW03 — Task 3 (Cross-Browser / Cross-Platform)
Assignment date:	28/07/2026
AI tool(s) used:	Claude Code (Claude Opus 5) — 1 orchestrator + 4 parallel sub-agents
AI tool(s) used:	☒ Yes ☐ No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GD1 · Platform matrix + environment declaration Prompt (verbatim — the decision exchange that produced this artifact): Question put to me: “Task 3 yêu cầu chạy checklist Task 1 trên ≥3 platform, screenshot phải hiện tên browser/OS/device + URL localhost. Máy này chỉ có Safari.app; Playwright đã cài sẵn Chromium/Firefox /WebKit. Anh muốn đi hướng nào?” — with 4 options offered (Playwright 3 engines local / Playwright + real Safari via <code>safaridriver</code> / BrowserStack–LambdaTest with my own credentials / 3 engines + mobile emulation). My decisions (verbatim):	<code>platform–matrix.md</code> : verified host facts (macOS 15.5 build 24F74, Apple Silicon, only <code>Safari.app</code> installed, no cloud trial), the 3 required platforms (Blink 151.0.7922.34 / Gecko 153.0 / WebKit 26.5, headed, 1280×800), the 2 emulated mobile profiles marked as excluded from the evidence set, and a 7-point honest-declaration section naming every substitution (Chrome-for-Testing instead of consumer Chrome; Playwright WebKit instead of <code>Safari.app</code> ; the Firefox bundle is <code>Nightly.app</code> ; P4/P5 are emulation, not devices).	INCOMPLETE	ISTQB FL – the test environment is part of the test basis: a result is only interpretable together with the environment that produced it, and §6 of this assignment makes the environment itself a graded artifact (“screenshots must clearly show the browser / OS / device name”). Choosing what to substitute for the unavailable cloud grid is a human risk decision, not an AI one.	I picked the 3-local-engine option after the trade-offs were laid out, supplied the identity to stamp, and cut the 2 emulated mobile platforms because emulation is not a real device and would only dilute the evidence set. I required §4–§5 of the file to state plainly that P3 is WebKit-not-<code>Safari.app</code> and that P4/P5 are emulation, plus the exact steps to close that gap later (<code>safaridriver --enable</code> , Expo Go on a real phone) — instead of letting the report imply “tested on Safari”.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
"Playwright 3 engine (local)" · overlay identity "dtnguyen23@clc.fitus.edu.vn" · later in the session "Bỏ 2 cái mobile web kia" → drop the two emulated mobile platforms.				

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GĐ2 · Harness core Prompt (verbatim, P1 — abridged here, full text in Appendix A): ROLE: Bạn dựng một harness Playwright để thực thi lại một checklist GUI đã có (66 item, Task 1) trên nhiều engine, không phải viết test mới. ... YÊU CẦU: 1. lib/platforms.js ... 2. lib/overlay.js : stamp platform + engine + version, OS, device, viewport + DPR, locale engine, URL thật, item ID, kết quả, thời điểm chạy, MSSV + họ tên + email ... 3. lib/ctx.js : goto , login (qua UI thật), addToCartFromHome , css/cssOf , snap , và tự động bắt + accept mọi alert/confirm/prompt ... 4. run-audit.js : N item × M platform, mỗi check một browser context mới, timeout 60 s/check, xuất	<pre>harness/lib/platforms.js , harness/lib/overlay.js , harness/lib/ctx.js , harness/run-audit.js + a 2-item smoke run on all three engines. The smoke run itself already produced a real cross-engine difference: Home price 30.000.000 VND on Blink/WebKit vs 30,000,000 VND on Gecko (engine locale vi / vi-VN / en-US).</pre>	INCOMPLETE	ISTQB FL – tooling & repeatability: 198 manual re-executions (66 × 3) are neither feasible in the time box nor free of confirmation bias, so the measurement must be automated and re-runnable; but a harness is a test artifact and must itself be verified before its output is trusted.	Two defects I found in the AI's first version and had fixed: (a) the overlay bar covered the SUT's own header, which would have destroyed the evidence for every IA-03 navigation item — fixed by shifting page content down by the bar height and restoring it after the shot; (b) page.screenshot() cannot capture browser chrome, so "URL visible in the picture" was only satisfied by the overlay text — I required the extra window-level capture path (row T3-GĐ6).

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
<pre>results/raw/<platform>.json ...</pre> RÀNG BUỘC: không dùng @playwright/test . Không hard-code kết quả Task 1.				
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GD3 · Check-module contract Prompt (verbatim, P2 — the contract itself, full text in Appendix A): 1. Observe, never assume. A check must read the live DOM / computed style / behaviour. Do not hard-code the Task 1 verdict. 2. Put every platform-visible raw value into metrics . The matrix builder diffs metrics across platforms; that diff is the actual Task 3 finding (e.g. thousandsSeparator: "," ".", validationMessage per engine). ... 8. BLOCKED is legitimate ... Explain why in evidence .	<pre>harness/checks/README.md : the return-value contract (PASS / FAIL / BLOCKED + evidence quoting the observed value + metrics carrying raw values), 8 rules, the full ctx API table, and the SUT seed facts (users, 5 products, 4 coupons, routes, cart-in-React-state).</pre>	VALID	ISTQB FL – test design: a shared, explicit oracle definition is what makes results from 4 independent implementers comparable. Forbidding hard-coded verdicts is the anti-confirmation-bias control that Task 3 needs, since the “expected” answers from Task 1 were already known to the implementers.	None required — but note the contract is <i>why</i> the disagreements with Task 1 surfaced at all (see the next four rows). I kept rule 1 non-negotiable when reviewing the modules: any check whose verdict could be produced without touching the app was rejected.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) — sub-agent 1 Date: 28/07/2026 Artifact: T3-GD3 · IA-01 checks (17 items) Prompt (verbatim, P3a — full text in Appendix A): ... DELIVERABLE: rewrite harness/checks/ia01.js ... EXACTLY these 17 checks ... DESIGN NOTES: colour items must read the real computed background-color ... IA01-13 (tab order) must actually press Tab and record the focused-element sequence ... IA01-08 ("never render NaN đ"): drive it through the UI (ctx.page.route to stub /api/products/1 with a non-numeric price) ... RULES: never hard-code the Task 1 verdict. It is expected (and fine) if some items disagree with Task 1.	checks/ia01.js (17 checks, ~1 035 lines). Findings: 16 FAIL / 1 PASS on every desktop engine; GUI-IA01-08 and GUI-IA01-15 disagree with Task 1's Passed; three real engine divergences recorded in metrics — money separator (vi / en-US / vi-VN), WebKit excluding <button> / <a> from the default Tab order, and 375 px horizontal overflow of the Home search row measuring 0 px (Blink) / 3 px (WebKit) / 24 px (Gecko).	INCOMPLETE	ISTQB FL – a test that only re-confirms a previous verdict adds no information; and test data selection determines defect detection: Task 1 exercised GUI-IA01-08 with valid seed data only, so it could not have seen NaN đ .	I adjudicated both disagreements myself instead of accepting them: IA01-08 FAIL is correct (a stubbed non-numeric price really renders NaN đ); IA01-15 FAIL is correct because at exactly 768 px Tailwind's md breakpoint is already active, so the grid is 3 columns where the checklist demands 2 — Task 1 had judged this by eye. Both are recorded as Task-1 corrections in the matrix, not silently overwritten.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) — sub-agent 2 Date: 28/07/2026 Artifact: T3-GĐ3 · IA-02 checks (15 items) Prompt (verbatim, P3b — full text in Appendix A): ... IA02-14 (required-message language): submit an empty required field and record the raw validationMessage string per engine — Chromium/Firefox /WebKit word it differently and may localise differently. Judge PASS only if the message is Vietnamese. ... IA02-09: besides reading the attributes, ACTUALLY type an invalid value, submit, and record input.validationMessage + input.validity flags in metrics .	checks/ia02.js (15 checks, ~1 088 lines). The headline Task 3 finding: GUI-IA02-14 flips verdict per engine — Blink "Vui lòng điền vào trường này." → PASS, Gecko "Please fill out this field." → FAIL, WebKit "Fill out this field" → FAIL, while navigator.language is vi-VN on both Blink and WebKit. Also GUI-IA02-09 : typing letters into type=number gives Gecko badInput=true + "Please enter a number.", whereas Blink/WebKit silently hand the app "" → parseInt("") = NaN .	INCOMPLETE	ISTQB FL – the test oracle must not be the implementation: relying on the browser’s built-in constraint-validation text means the “Vietnamese UI” requirement is satisfied or violated by the <i>user’s browser</i>, not by the SUT. This is the textbook case for cross-browser testing rather than single-browser testing.	I checked this by hand on all three engines before accepting it, because a verdict that flips per platform is exactly what an over-eager harness would fake. It reproduces. I then decided its classification myself: the engine wording difference is not an engine bug — the defect belongs to the SUT (it must render its own Vietnamese validation message instead of delegating to HTML5 required), so it is filed as a SUT bug whose <i>visibility</i> is platform-dependent.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) — sub-agent 3 Date: 28/07/2026 Artifact: T3-GD3 · IA-03 checks (15 items) Prompt (verbatim, P3c — full text in Appendix A): ... IA03-10, IA03-11 and IA03-09 need real <code>page.goBack()</code> / <code>page.goForward()</code> calls. Record the URL and the visible heading BEFORE and AFTER each history action in <code>metrics</code> — Chromium/Firefox /WebKit differ in bfcache/state restoration for a React SPA whose step lives in component state. ... IA03-07: detect a full document reload, not just the URL change — set a marker on <code>window</code> ... also count <code>framenavigated</code> / <code>load</code> events.	<code>checks/ia03.js</code> (15 checks, ~659 lines). 12 FAIL / 3 PASS — reproduces Task 1 exactly on all three desktop engines. Divergences: sub-pixel grid columns differ on all three engines (<code>314.664</code> / <code>314.667</code> / <code>314.65625...</code> <code>314.671875</code> px); Gecko reports 3 <code>load</code> events where Blink/WebKit report 2. Also recorded, though outside the item's wording: the order is re-submittable via <i>Forward</i> on every engine (<code>successFlagRestoredOnForward: false</code>).	INCOMPLETE	ISTQB FL – regression risk of environment change: the checklist item only judges <i>Back</i>, so the Forward-resubmit observation is a new test condition discovered during execution and must be reported rather than folded into an existing verdict.	I accepted the 12/3 split (it matches my Task 1 manual result) and kept the Forward-resubmit observation as a separate finding instead of letting it change <code>GUI-IA03-10</code> 's verdict. I also asked for <code>GUI-IA03-15</code> 's <code>observationLimit</code> metric to state explicitly that only 5 seeded products existed, so nobody reads that PASS as “pagination verified at scale”.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) — sub-agent 4 Date: 28/07/2026 Artifact: T3-GĐ3 · IA-04 checks (19 items) Prompt (verbatim, P3d — full text in Appendix A): ... IA04-12 (coupon feedback both branches, amounts correct) ... Verify the arithmetic against the rendered numbers — beware that thousands separators differ per engine, so strip non-digits before comparing. ... IA04-11: use a THROWAWAY account you register yourself so you never lock <code>test@eshop.com</code> ... IA04-13: inject <code></code> style payloads and detect execution WITHOUT breaking the harness.	<code>checks/ia04.js</code> (19 checks, ~948 lines). 19 FAIL on every engine. GUI-IA04-12 disagrees with Task 1's Passed: both coupon branches do render, but a 10 % <code>SAVE10</code> on 30 000 000 đ shows <code>Tiết kiệm -270.000.000 đ</code> and <code>Thành tiền 300.000.000 đ</code> — the backend computes <code>discount = total × (1 - discount_value)</code>. Divergence: how a swallowed 500 is observable — Blink logs 4 console errors incl. status code + stack, WebKit the same 2 messages without stack, Gecko only bare <code>"Error" / "Lỗi lấy đơn hàng: Error"</code>. Non-divergence, deliberately instrumented: <code>alert()</code> and the <code></code> payload behave identically on all three.	INCOMPLETE	ISTQB FL – a checklist item is a test condition, not a click sequence: <code>GUI-IA04-12</code> demands “cà 2 nhánh đúng, số chính xác”, so verifying only that both feedback branches appear leaves the stated condition partially untested.	I re-derived the coupon arithmetic by hand (10 % of 30 000 000 = 3 000 000, expected total 27 000 000) and re-ran the flow myself before accepting that Task 1's Passed was wrong; this is the most valuable single finding of Task 3 and it is a functional bug, not a platform one, so I filed it as such and did not dress it up as a cross-platform issue.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GĐ4-5 . Authoritative run + matrix/divergence generation Prompt (verbatim, P4): Từ <code>results/raw/*.json</code>, sinh <code>results-matrix.md</code> (bảng 66 item x platform + tổng hợp theo platform và theo aspect + bảng đối chiếu với kết luận Task 1) và <code>divergences.md</code> chia 2 phần: (A) item đổi Pass/Fail giữa các platform — kèm bảng giá trị <code>metrics</code> khác nhau và link screenshot từng platform; (B) item cùng kết quả nhưng giá trị hiển thị khác nhau. Sinh bằng script (<code>scripts/build-matrix.js</code>) để chạy lại được, không viết tay.	<pre>harness/scripts/build-matrix.js + generated results-matrix.md and divergences.md ; harness/run-all-platforms.sh executes the 3 required platforms with a database re-seed before each one.</pre>	INCOMPLETE	ISTQB FL – test data management: results from runs whose starting state differs are not comparable. Several checks legitimately mutate state (they place real orders, register users, edit the profile phone) because that is what the Task 1 steps require.	I reversed the AI's own procedural decision here. Its first plan said <i>not</i> to restart the backend between platforms; the evidence said otherwise — <code>GUI-IA03-15.orderRows</code> drifted 7/3/5/9/11 with run order. Since <code>backend/databases.js</code> drops and re-seeds on start, the authoritative run now re-seeds before every platform, and the retracted instruction is documented as a mistake in <code>plan/PLAN-TASK3.md</code> §GĐ0 instead of being quietly deleted.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GĐ6 · Window-level evidence capture Prompt (verbatim, P5): Playwright chỉ chụp được viewport, không chụp được browser chrome. Viết <pre>scripts/capture-platform-proof.js : với mỗi platform, mở 6 màn đại diện ..., stamp overlay email/engine/OS/device, đưa cửa sổ lên trước, đọc window.screenX/screenY/outerWidth/outerHeight rồi gọi screencapture -R để chụp đúng vùng cửa sổ browser + menu bar macOS — không chụp cả desktop (tránh lọt nội dung riêng tư của máy).</pre>	<pre>harness/scripts/capture-platform-proof.js → results/<platform>/platform-proof/*.png : real browser windows showing the macOS menu bar (browser app name) and the browser's own address bar with localhost:5173/... , plus the in-page overlay with my student e-mail.</pre>	INCOMPLETE	§6 of the assignment ("screenshots clearly show the browser / OS / device name alongside the SUT's localhost URL") and §11 (screenshots must not be fabricated): the strongest satisfying evidence is an unedited OS-level capture of the real window, and privacy requires cropping to that window rather than grabbing the whole screen.	Three iterations were needed and I checked every image by eye: the first attempt tried AppleScript/System Events and was refused (no Accessibility permission), the second captured a <i>different application's</i> window that happened to be in front of the WebKit window, and only the geometry read from inside the page (<code>window.screenX/screenY/outerWidth/outerHeight</code>) produced correct crops. I also required <code>--disable-features=TranslateUI</code> after Chrome's translate popup appeared over the page in the first capture.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Opus 5) Date: 28/07/2026 Artifact: T3-GĐ7 · Evidence gate + report assembly Prompt (verbatim, P6): Viết <code>scripts/verify-evidence.js</code> chặn deliverable nếu: platform nào thiếu item nào so với 66 ID trong <code>checklist-final.md</code> ; item FAIL/ERROR nào không có file ảnh thật trên đĩa; còn item nào ở trạng thái ERROR; ảnh nào < 5 KB; platform thật nào chưa có ảnh cửa sổ. Fail thì exit 1 kèm danh sách vấn đề.	<code>harness/scripts/verify-evidence.js</code> (exits non-zero with the problem list) + <code>report.md</code> , <code>issues/XP-*.md</code> .	INCOMPLETE	ISTQB FL – test completion criteria must be checkable, not asserted: without a gate, a partially-completed run reads exactly like a complete one in the final report.	I required the gate to cross-check against the <i>source</i> checklist (<code>checklist-final.md</code>), not against the harness's own list of checks — otherwise a check silently dropped from a module would also disappear from the expectation. The final classification of every divergence (SUT bug / legitimate engine behaviour / measurement limitation) in the report is mine, and the AI Critique is written by me.

4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	10	100%
VALID (correct, accepted as-is)	1	10%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	9	90%

5. Conclusion — When should AI be used (or not)?

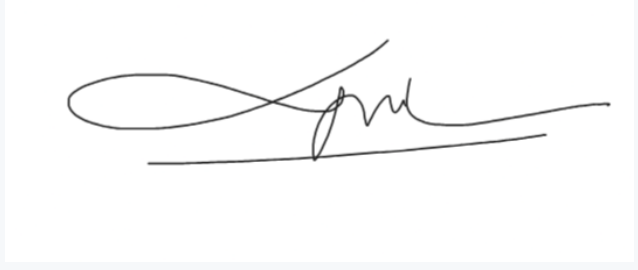
Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

AI was the only realistic way to re-execute 66 checklist items on 3 platforms (198 executions) inside the time box, and it was excellent at exactly the part humans do worst: recording the raw value behind every verdict, which is what turned “all three browsers look the same” into 29 measured divergences (1 that flips the verdict, 28 in the displayed values). It was weakest at the meta-decisions — the evidence layer it first produced could not show a URL bar at all, its overlay hid the very header some items were about, and its own run procedure would have polluted the comparison it was designed to make. Rule I take away: let AI build the instrument and read the dials, but never let it decide what counts as a valid measurement, and always ask “what would this look like if it were wrong?” before believing a verdict that flips per platform.

6. Mandatory Disclosure (paste verbatim)

For Task 3, Claude Code designed and wrote the entire measurement harness (platform definitions, evidence overlay, per-check context, runner, matrix/divergence generator, window-capture script, evidence gate), implemented all 66 checks from my Task 1 test cases, executed them on the 3 required platforms (198 executions), and drafted the matrix, divergence list and report. I chose the platform matrix and accepted its documented risks, reversed the AI’s re-seeding procedure after the metrics showed run-order pollution, re-verified by hand every item whose result contradicted Task 1 (GUI-IA01-08 , GUI-IA01-15 , GUI-IA04-12) and the one item whose verdict flips per engine (GUI-IA02-14), checked every window screenshot by eye, and made the final call on which divergences are SUT bugs versus legitimate engine behaviour. **No screenshot in this submission was generated, drawn or retouched by AI:** every image is an unedited capture from a real browser run on my own machine, with the overlay rendered by the browser itself before the shot. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

Student name (printed):	DANG TRUONG NGUYEN
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Class / Cohort:	23KTPM3
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Msc. Tran Thi Bich Hanh
Date:	28/07/2026
Signature:	

References

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